ANSURES EXTENDED RELEASE TABLET
Metformin (as hydrochloride) 500 mg

Description
White tablets, oblong with length 17.6 mm $\pm$ 0.25mm, width 8.0mm $\pm$0.25mm, thickness 6.65mm $\pm$ 0.15mm.

Composition
Each tablet contains 500mg of Metformin (as hydrochloride) 500mg.

Action
Metformin is an antihyperglycemic agent which improves glucose tolerance in patients with type 2 diabetes, lowering both basal and postprandial plasma glucose. Its pharmacologic mechanisms of action are different from other classes of oral antihyperglycemic agents. Metformin decreases hepatic glucose production, decreases intestinal absorption of glucose, and improves insulin sensitivity by increasing peripheral glucose uptake and utilization. Unlike sulfonylureas, metformin does not produce hypoglycemia in either patients with type 2 diabetes or normal subjects (except in special circumstances) and does not cause hyperinsulinemia.
With metformin therapy, insulin secretion remains unchanged while fasting insulin levels and day-long plasma insulin response may actually decrease.

Summary of pharmacodynamics and pharmacokinetics
Pharmacology: Metformin is a biguanide with antihyperglycaemic effects, lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycaemia.

Metformin may act via 3 mechanisms: Reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis; in muscle, by increasing insulin sensitivity, improving peripheral glucose uptake and utilisation; and delay of intestinal glucose absorption.

Metformin stimulates intracellular glycogen synthesis by acting on glycogen synthase.

Metformin increases the transport capacity of all types of membrane glucose transporters (GLUT).

In humans, independently of its action on glycaemia, immediate-release metformin has favourable effects on lipid metabolism. This has been shown at therapeutic doses in controlled, medium-term or long-term clinical studies: Immediate-release metformin reduces total cholesterol, LDL cholesterol and triglyceride levels.

For metformin used as 2nd-line therapy, in combination with a sulphonylurea, benefit regarding clinical outcome has not been shown.

In type 1 diabetes, the combination of metformin and insulin has been used in selected patients, but the clinical benefit of this combination has not been formally established.

Pharmacokinetics: Absorption: After an oral dose of the extended-release tablet, metformin absorption is significantly delayed compared to the immediate-release tablet with a T_{max} at 7 hrs (T_{max} for the immediate-release tablet is 2.5 hrs).

At steady-state, similar to the immediate-release formulation, C_{max} and AUC are not proportionally increased to the administered dose. The AUC after a single oral administration of 2000 mg of metformin extended-release tablets is similar to that observed after administration of 1000 mg of metformin immediate-release tablets twice a day.

Intrasubject variability of C_{max} and AUC of metformin extended-release is comparable to that observed with metformin immediate-release tablets.

Although the AUC is decreased by 30% when the extended-release tablet is administered in fasting conditions, both C_{max} and T_{max} are unaffected.

Metformin absorption from the extended-release formulation is not altered by meal composition.

No accumulation is observed after repeated administration of up to 2000 mg of metformin as extended-release tablets.

Distribution: Plasma protein-binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean V_d ranged between 63-276 L.

Metabolism: Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination: Renal clearance of metformin is >400 mL/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hrs.

When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus, the elimination half-life is prolonged, leading to increased levels of metformin in plasma.

Toxicology: Preclinical Safety Data: Preclinical data reveal no special hazard for humans based on conventional studies on safety, pharmacology, repeated-dose toxicity, genotoxicity, carcinogenic potential and toxicity reproduction.

Indications

Metformin Hydrochloride used for treatment of type 2 diabetes mellitus in adults, when dietary management and exercise alone does not result in adequate glycaemic control. Metformin extended release may be used as monotherapy or in combination with other oral antidiabetic agents, or with insulin.

Recommended Dosage

Monotherapy and combination with another oral antidiabetic agents:

- The usual starting dose is one tablet once daily.

- After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastro-intestinal tolerability. The maximum recommended dose is 4 tablets daily.
- Dosage increases should be made in increments of 500mg every 10 to 15 days, up to maximum of 2000mg once daily with the evening meal.
- In patients already treated with metformin tablets, the starting dose of Metformin 500mg extended release should be equivalent to the daily dose of metformin immediate release tablets.
- If transfer from another oral antidiabetic agent is intended: discontinue the other agent and initiate metformin 500mg extended release at the dose indicated above.

Combination with insulin:
Metformin and insulin may be used in combination therapy to achieve better blood glucose control. The usual starting dose of metformin 500mg extended release is one tablet once daily, while insulin dosage is adjusted on the basis of blood glucose measurements.

***Elderly*:**
due to the potential for decreased renal function in elderly subjects, the metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary.

Children: In the absence of available data, metformin 500mg extended release should not be used in children.

Renal impairment
A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR ml/min	Total maximum daily dose	Additional considerations
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
< 30	-	Metformin is contraindicated.

Contraindications
Metformin Hydrochloride is contraindicated in patients with hypersensitivity to Metformin Hydrochloride or other of inacative ingredient. It also contraindicated in patients with diabetic ketoacidosis, diabetic pre-coma, renal failure or renal dysfunction (creatinine clearance<60mL/min). Acute conditions with the potential to alter renal function such as cardiac or respiratory failure, recent myocardial infarction and shock). Metformin Hydrochloride contraindicated in patient with hepatic insufficiency, acute alcohol intoxication, alcoholism and lactating mothers. Metformin also contraindicated in

- Severely reduced kidney function (GFR < 30mL/min)
- Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)

Warning and Precautions
Lactic acidosis:
Lactic acidosis is a rare, but serious (high mortality in the absence of prompt treatment), metabolic complication that can occur due to metformin accumulation. Reported cases of lactic acidosis in patients on metformin have occurred primarily in diabetic patients with significant renal failure. The incidence of

lactic acidosis can and should be reduced by assessing also other associated risk factors such as poorly controlled diabetes, ketosis, prolonged fasting, excessive alcohol intake, hepatic insufficiency and any condition associated with hypoxia.
Diagnosis: Lactic acidosis is characterized by acidotic dyspnoea, abdominal pain and hypothermia followed by coma. Diagnostic laboratory findings are decreased blood pH, plasma lactate levels above 5 mmol/L, and increased anion gap and lactate/pyruvate ratio. If metabolic acidosis is suspected, metformin should be discontinued and the patient should be hospitalized immediately.
Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.
In case of dehydration (severe diarrhea or vomiting, fever or reduces fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.
Medical products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.
Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterized by acidotic dyspnea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (<7.35), increased plasma lactate levels (>5mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal Function:
As metformin is excreted by the kidney, creatine clearance (this can be estimated using the Cockcroft-Gault formula) and /or serum creatinine levels should be determined before initiating treatment and regularly thereafter:

- At least annually in patients with normal renal function,
- At least two to four times a year in patients with serum creatinine levels at the upper limit of normal and in elderly subjects.

Decreased renal function in elderly subjects is frequent and asymptomatic. Special causion should be exercised in situations where renal function may become impaired, for example when initiating antihypertensive therapy or diuretic therapy and when starting therapy with an NSAID.
GFR should be assessed before treatment initiation and regularly there after [See Section Recommended Dosage]. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function [See Section Contraindications].

Administration of iodined contras agent:
As the intravascular administration of iodinated contrast materials in radiologic studies can lead to renal failure, metformin should be discontinued prior to, or at the time of the test and not reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Surgery:
Metformin hydrochloride should be discontinued 48 hours before elective surgery with general anaesthesia and should not be usually resumed earlier than 48 hours afterwards.

Other precautions:

- All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet.
- The usual laboratory tests for diabetes monitoring should be performed regularly.
- Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or sulphonylureas.
- The tablet shells may be present in the faeces. Patients should be advised that this is normal.

Interaction with Other Medicaments

Nifedipine enhances the absorption of Metformin. Cationic drugs like Cimetidine. Amiloride, Digoxin, Morphine, Procainamide, Quinidine, Quinine, Triamterene, Trimethoprim and Vancomycin that are eliminated by renal tubular secretion have potential for interaction with Metformin by competing for common renal tubular transport systems. The patient should be closely observe for loss of blood glucose control when thiazides and other diuretics, corticosteroids, phenothiazines, thyroid products, estrogens, oral contraceptives, phenytoin, nicotinic acid, sympathomimetics, calcium channel blocking drugs and isoniazid are administered together with Metformin.

Pregnancy and Lactation

Category B: Either animal-reproduction studies have not demonstrated a foetal risk but there are no controlled studies in pregnant women or animal-reproduction studies have shown an adverse effect (other than a decrease in fertility) that was not confirmed in controlled studies in women in the 1st trimester (and there is no evidence of a risk in later trimesters).

Side Effects

The following undesirable effects may occur under treatment with metformin. Frequencies are defined as follows: very common:>1/10
Common: >1/100,1/10
Uncommon:>1/1000, <1/100
Rare >1/10,000
Very rare:<1/10,000

<u>Metabolism and nutrition disorders:</u>

Very rare: Decrease of vitamin B_{12} absorption with decrease of serum levels during long-term use of metformin. Consideration of such aetiology is recommended if a patient presents with megaloblastic anaemia.

Very rare: lactic acidosis

<u>Nervous system disorders:</u>

Common: Taste disturbance

<u>Gastrointestinal disorders:</u>

Very common: gastrointestinal disorders such as nausea, vomiting, diarrhea, abdominal pain and loss of appetite. These undersirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. A slow increase of the dose may also improve gastrointestinal tolerability.

<u>Hepatobiliary disorders:</u>

Isolated reports: Liver function tests abnormalities or hepatitis resolving upon metformin discontinuation.

Skin and subcutaneous tissue disorders:
Very rare: Skin reactions such as erythema, pruritus, urticaria.

Symptoms and Treatment of Overdose
Hypoglycaemia has not been seen with metformin doses of up to 85 g, although lactic acidosis has occurred in such circumstances. High overdose or concomitant risks of metformin may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in the hospital. The most effective method to remove lactate and metformin is haemodialysis.

Presentation
This tablet is available in 28's and 56's
14 tablets are packed into one blister

Storage conditions
Protect from light and store below 30^{0}C. Keep out of reach of children.

Shelf-life
The tablets can be used within 48 months from the date of manufacture if kept as recommended.

Registration Number
ANSURES EXTENDED RELEASE TABLET – MAL10080062AZ

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